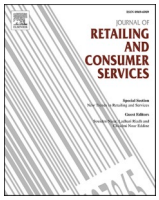
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How emotions in online reviews affect movie sales: Evidence from Hollywood

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ABSTRACT

Movies evoke strong human emotions that resonate in the content of online reviews, known as electronic word of mouth (*e*WOM). Existing literature suggests that positive content enhances, and negative content reduces movie sales, with negative content having a stronger effect. We demonstrate that there is a limit to the negative emotional content's impact on sales, beyond which it boosts sales. We extract positive and negative emotional content from 80 Hollywood movie reviews ($N = 23{,}046$) and empirically examine their influence on movie sales. Results indicate that for high *e*WOM volume, the effect of positive content increases progressively, while negative content has diminishing returns. Grounded in 'Emotions as Social Information' (EASI) theory, this study provides marketers and researchers with a new understanding of how emotional content in *e*WOM influences moviegoers' choice of movie and sales. Our typology—Flop, Hype, Star, and Hit—helps predict movie sales and design targeted strategies.

1. Introduction

Online media provides a vibrant platform for consumers to publicize their positive and negative reviews about a product. These reviews, known as electronic word of mouth (*e*WOM), are a powerful source of first-hand information that profoundly influences customers' buying behavior (Babić Rosario et al., 2016; Floyd et al., 2014; Lee et al., 2017; Kong and Lou, 2023; Li et al., 2020; Yim et al., 2023). Since *e*WOM spreads seamlessly through online media, it can significantly boost or decline sales (Floyd et al., 2014; Li et al., 2020; Sun, 2011; Yim et al., 2023). Concerned with this notion, one of the most prolific streams of *e*WOM research has focused on the movie market (Chintagunta et al., 2010; Lee et al., 2017; Goldenberg and Willer, 2023). Movies, due to their experiential nature and short lifespan, rely heavily on *e*WOM for their success or failure (Eliashberg et al., 2006; Liu, 2006). Existing studies used three key metrics to examine online movie reviews: *volume*—the quantum of reviews, *valence*—the quality of reviews as negative or positive, and *variance*—derived measure from the valence showing heterogeneity in consumer ratings (Basuroy et al., 2020; Fan et al., 2021; Lee et al., 2017; Liu, 2006; Ullah et al., 2016).[1] Although most research has concluded that *e*WOM valence (positive vs. negative) impacts movie sales beyond other marketing mix effects (Chintagunta et al., 2010; Rosario et al., 2016), the findings are inconclusive. Some studies found a significant impact of *e*WOM valence and movie sales (Chintagunta et al., 2010; Robertson et al., 2023), conversely, others have shown no meaningful relationship (Fan et al., 2021; Rocklage and Fazio, 2020). Moreover, while many have noted that negative *e*WOM decreases sales (Chevalier and Mayzlin, 2016; Tsao, 2014), others argue that negative *e*WOM increases product evaluations and can enhance sales (e.g., Hiura et al., 2010; Kikumori and Ono, 2013; Meeker, 2022). These varying results hinder marketers from making informed decisions about *e*WOM management. We argue that one of the main causes of this variance is the use of sentiments as a proxy for emotions. Prior studies have mostly relied on positive and negative sentiments (Hu et al., 2014; Lee et al., 2017), yet positive and negative emotional content, which greatly drives *e*WOM (Goldenberg and Willer, 2023), remained understudied.

It is vital to note that emotions differ markedly from sentiments, exerting a unique influence on consumer decision-making and purchase likelihood (Yu et al., 2023). Sociological theories, such as 'Emotions as

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[1] Variance as compared to volume and valence, reflects a more unique and intriguing product such as more niche categories (Sun, 2012), and have limited visibility (Babić Rosario et al., 2016). Therefore, variance owes its visibility to signalling more interestingness (Xi, 2018) rather than by serving as a proxy for the quality of experience attributes.

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Social Information' (EASI), provide insight into the mechanism by which emotional content in *e*WOM influences consumers' affective reactions, inferential processes, and buying behavior (Van Kleef et al., 2011; Van Kleef, 2017). Unlike sentiments that develop over time as individuals continuously reflect on specific objects to form a stable favorable or unfavorable mindset (Ullah et al., 2016), emotions are transient affective states of mind that evoke immediate arousal (Berger, 2014; Septianto and Chiew, 2018). Accordingly, sentiments have a more gradual and sustained impact on consumer behavior (Ozen and Engizek, 2014), but emotions such as frustration or anger have an impulsive and powerful impact on buying behavior (Guo et al., 2020), with consumers sometimes overlooking the financial consequences of their purchases (Ozen and Engizek, 2014; Sharma et al., 2010). In addition, according to EASI theory, social consensus influences consumers' affective reactions (Van Kleef, 2009), suggesting that *e*WOM volume, as an indicator of social consensus, can play a vital role in persuading consumers. Prior studies have highlighted the important role of *e*WOM volume in movie performance (Carrillat et al., 2018; Duan et al., 2008; Lee et al., 2017). Given the distinctive characteristics of emotional content and its close association with the spread of *e*WOM (Ullah et al., 2016), it is important to study the specific impact of positive and negative emotional content on movie sales and examine the role of *e*WOM volume in this relationship. To our knowledge, this analysis has not been contemplated in previous research.

This study aims to achieve two main objectives. First, we use EASI theory (Van Kleef, 2009) to study the specific impact of positive and negative emotional content on movie sales. While studies across various domains, including retailing and consumer service, have increasingly used EASI theory to understand how consumers use emotional expressions as a powerful source of social information and how this might influence their behavior (Ismagilova et al., 2021; Urbonavicius, 2023; Zhang et al., 2024), this lens has not been used to predict movie sales. Second, we investigate the moderating role of eWOM volume in the above relationship because prior studies suggest that increasing *e*WOM volume can play an important role in persuading others (Chevalier and Mayzlin, 2016; Babić Rosario et al., 2016; Duan et al., 2008; Hu et al., 2019; Lee et al., 2017). This is particularly pertinent to movie reviews, which usually have a wide range of positive and negative emotional content (Ullah et al., 2016). Our data comprising 23,046 reviews about 80 Hollywood movies show that while positive and negative emotional content in *e*WOM directly affects product sales, the impact of negative emotional content is significantly stronger. However, with the rising *e*WOM volume, the positive emotional content exhibits a linear increase in movie sales, whereas the negative emotional content has a diminishing effect, i.e., negative emotional content exerts a stronger impact when *e*WOM volume is low, but this effect weakens as the *e*WOM volume escalates.

The study makes three key contributions. First, it advances the eWOM research stream by examining the impact of positive and negative emotional content on movie sales separately from sentiments, thus providing empirically supported guidance regarding *how and to what extent positive and negative emotional content in eWOM impacts movie sales*. Second, it develops a 2x2 matrix by integrating emotional content valence and volume, which complements the EASI theory (Van Kleef, 2009) and contributes to a better understanding of how positive and negative emotional content in *e*WOM shapes human buying behavior for experience products (movies). Finally, it responds to calls from researchers to examine why expressions of emotions on social media wrap collective attitude (Goldenberg and Willer, 2023), how emotional content in eWOM can be used to predict consumer decision-making (Rocklage and Fazio, 2020), and what is the impact of positive and negative emotions on product performance (Fan et al., 2021; Robertson et al., 2023; Yu et al., 2023).

2. Theory and hypotheses

The EASI theory provides a succinct and conclusive framework to understand how emotional content in online reviews can influence human buying behavior. Rooted in a social-functional approach to emotion (Van Kleef, 2017; Parkinson, 1996), this theory posits that "emotional expressions provide information to observers, which may influence their behavior" (Van Kleef, 2009, p. 184). The EASI model expands this notion further by identifying two important processes through which emotional expressions influence observers: a) inferential process, and b) affective reactions (Alkhadher et al., 2023; Deng et al., 2020; Van Kleef, 2017). *Inferential process*—observers infer information from others' emotional expressions, e.g., information about their attitudes, feelings, behavioral intentions, and relational orientation (Keltner and Haidt, 1999; Van Kleef, 2017). Such inferences, in turn, influence the observer's behavior (Van Kleef, 2009). *Affective process*—emotional expressions elicit normative or affective reactions in observers, which subsequently affect their behavior (Alkhadher et al., 2023; Deng et al., 2020). The theory further argues that such affective reactions can emerge in two ways. First, emotions are directly transmitted from an expresser to the observer via emotional contagion, which can involve mimicry, mirror-neuron activity, and afferent feedback (i.e., from verbal, facial, and postural expressions). Second, emotional expressions create impressions, affecting interpersonal liking and social intentions. In other words, emotional expressions are a source of social influence (Van Kleef et al., 2011), and can persuade others in a particular direction, particularly when they are expressed by many people (Sridhar and Srinivasan, 2012).

It is well-documented that emotions trigger *e*WOM loaded with a higher proportion of positive and negative emotional content (Barta et al., 2023; Septianto and Chiew, 2018; Ullah et al., 2016), which have greater virality in comparison to neutral words (Berger, 2014; Berger and Milkman, 2012; Berger et al., 2021). By drawing on EASI theory, the impact of positive and negative emotional content on consumer behavior can be studied at a more granular level.

2.1. Effects of positive and negative emotional content on movie sales

According to EASI theory (Van Kleef, 2017), emotional expressions wield interpersonal influence by triggering an inferential process. The positive and negative emotional content in *e*WOM will trigger emotional reactions in observers (moviegoers) and subsequently influence their behavior. Research shows that movie reviews contain a higher proportion of positive and negative emotional content (Barta et al., 2023; Robertson et al., 2023; Ullah et al., 2016), and emotional content is shared more frequently than neutral content (Berger and Milkman, 2012; Stieglitz and Dang-Xuan, 2013). The EASI theory suggests that consumers infer information from emotional content, such that positive/negative emotional expression shows a positive/negative favorable appraisal (Van Kleef and De Dreu, 2010), which affects buying behavior.

By human nature, consumers prefer sharing positive content that uplifts others' moods or provides information about rewarding experiences (Palmeira et al., 2020). This tendency is driven by the desire to be seen as someone who contributes positively rather than someone who conveys negativity or sadness (Palmeira et al., 2020). Sharing positive reviews serves both self-presentation and identity communication purposes (Wojnicki and Godes, 2017). Because positive emotional content reflects favorably on the sender, it is disseminated more (Robertson et al., 2023). The seminal work of Berger and Milkman (2012) revealed that positive emotional content is circulated more widely compared to negative emotional content, as evidenced by the analysis of nearly 7000 articles from The New York Times. Furthermore, many studies in the past, including those by Humphreys (2010) and Packard and Berger (2017) have reported that positive emotions are correlated with known positive attributes and consumption. In this regard, satisfied moviegoers

who experience overwhelming entertainment are likely to transmit positive emotional words in their online reviews (Robertson et al., 2023). The positive attributes of the movie in *e*WOM, such as an engaging storyline, exceptional acting, stunning visuals, and relatable characters, contribute to positive consumption utility (Malik and Hussain, 2017), leading to higher sales. Thus, we hypothesize:

H1. Positive emotional content in *e*WOM will have a positive impact on movie sales.

On the other hand, dissatisfied consumers, as indicated by previous research, frequently engage in a process known as dissonance reduction—expressing negative emotional content about a movie in their online reviews, a means through which they convey discontent to alleviate frustration, tension, and anxiety (Dellarocas and Narayan, 2006; Manthiuo et al., 2020; Robertson et al., 2023; Ullah et al., 2015). These expressions of negative emotions in *e*WOM can adversely impact movie performance. Prior studies (e.g., Berger and Milkman, 2012; Sung et al., 2023) reported that negative emotions damage brand reputation and perceptions of product quality. Negative emotions in online reviews, including anger, fear, and sadness, decrease the overall attitude toward a product (Zablocki et al., 2019). In the context of *e*WOM, Berger et al. (2010) and Robertson et al. (2023), noted that negative emotional words damage product evaluations and reduce the purchase likelihood.[2]

H2. Negative emotional content in *e*WOM will have a negative impact on movie sales.

Despite the higher frequency of positive emotional content in online movie reviews (Wojnicki and Godes, 2017; Ullah et al., 2015), research indicates that negative content carries a more substantial impact (Kong and Lou, 2023; Tsao, 2014). This phenomenon can be attributed to the basic human tendency to give more attention to negative expressions, a notion well-established in human psychology (e.g., Baumeister et al., 2001). Expanding on this notion, Robertson et al. (2023) uncovered a strong relationship between negative words in *e*WOM and user engagement. Specifically, they reported that for each additional negative word in a review, there is a notable 2.3 % increase in the click-through rate. This observation is complemented by findings from Xu (2020), who identified that consumers experiencing negative emotions tend to provide stronger and more detailed comments. In line with the ESAI theory (Van Kleef, 2017), compared to positive emotional content, negative emotional content in *e*WOM will carry greater inferential power to sway moviegoers and impact their purchase decisions. Since negative information gets more attention than positive and neutral information (Brady et al., 2023; Goldenberg and Willer, 2023), it becomes more informative, attention-grabbing, and carries more weight in purchase decisions (Sen and Lerman, 2007; Tsao, 2014). Consumers tend to place more weight on negative versus positive information, resulting in "negativity bias" or "negativity effect" (Ahmad and Guzmán, 2021; Kanouse, 1984; Tata et al., 2020). Therefore:

H3. Negative emotional content in *e*WOM will have a stronger impact on movie sales than positive emotional content.

2.2. Moderating role of eWOM volume

We expect popularity and majority opinion to be a contingency factor in the above-alluded relationships. Previous research shows that consumers infer the correctness of opinions from their popularity (Rosario et al., 2016; Salganik et al., 2006). According to the affective aspect of EASI theory (Van Kleef, 2017), people tend to conform to normative behavior, aligning with others' positive expectations. Consumers shape their opinions about a product based on the prevailing popularity or majority consensus (Hu et al., 2019). In the context of *e*WOM, studies highlight the bandwagon effect whereby consumers reduce uncertainty about a product's quality by following the choices of others.

A high *e*WOM volume indicates popularity and greater interest (Rosario et al., 2016; Salganik et al., 2006; Duan et al., 2008), enhancing the social influence of positive emotional content. This notion is based on the following reasons. Firstly, when consumers observe a rising number of positive reviews, they are more likely to be swayed by the bandwagon effect (Chevalier and Mayzlin, 2016; Babić Rosario et al., 2016), which reinforces their confidence in watching the movie (Carrillat et al., 2018). Secondly, drawing from EASI theory, when consumers encounter popular positive emotional content, they tend to rely less on information processing that increases the predictive power of affective reactions (Van Kleef, 2009, 2017)—high volume and positive emotional reviews act as favorable heuristic cues, indicating popularity and quality. Hence, owing to the affective reactions and bandwagon effect (driven by perceived popularity and consensus), the high *e*WOM volume enhances the impact of positive emotional content on movie sales. Thus:

H4a. High *e*WOM volume will amplify the impact of positive emotional content on movie sales.

Contrastingly, high *e*WOM volume can attenuate the impact of negative emotional content exhibiting a diminishing return. This assumption rests on the following reasons: firstly, volume as the total quantity of eWOM of reviews, whether dominated by positive reviews or with mixed reviews, creates a sense of popularity and interest, which consumers perceive as a positive cue for evaluating product quality (Babić Rosario et al., 2016). According to EASI theory, when consumers encounter a high volume and negative emotional content, they will engage in thorough information processing rather than relying on heuristic cues (Van Kleef, 2017). EASI theory further posits, "the more thorough the information processing, the stronger the predictive power of inferences" (Van Kleef, 2009, p. 186). This thorough information processing can dilute the social influence of negative emotional content and diminish its negative impact on sales. Secondly, counter *e*WOM from people who identify strongly with the focal movie will come into play. The growing number of negative *e*WOM may seem like a threat to their identity (Meeker, 2022), resulting in intergroup hostility, leading to counter-reviews and a bandwagon effect. The opposite reviews can weaken the impact of negative emotional content (Goldenberg and Willer, 2023; Van Kleef and De Dreu, 2010). As Brady et al. (2023: 918) argue, "perceiving the emotions of many group members at once makes people over-perceive the extremity of group emotions." In conforming to their group opinion, members start to post their reviews, which persuade potential moviegoers' preference for the movie and influence their purchase intention (Meeker, 2022). Marketing research indicates that the effects of negative *e*WOM on consumer attitude towards a product can diminish when it is followed by positive reviews (Hiura et al., 2010; Kikumori and Ono, 2013). In sum, thorough information processing and the bandwagon effect counter reviews can attenuate the adverse effect of negative emotional content on movie sales, suggesting that:

H4b. High *e*WOM volume will weaken the impact of negative emotional content on movie sales.

The hypothesized model is presented in Fig. 1.

3. Method

3.1. Sample

Movies are important tools for studying emotional stimuli generated

[2] Because literature has extensively covered the impact of positive and negative content on consumer behavior, we only briefly described them to hypothesize the main effects and move to the more novel effects that have received scant attention.

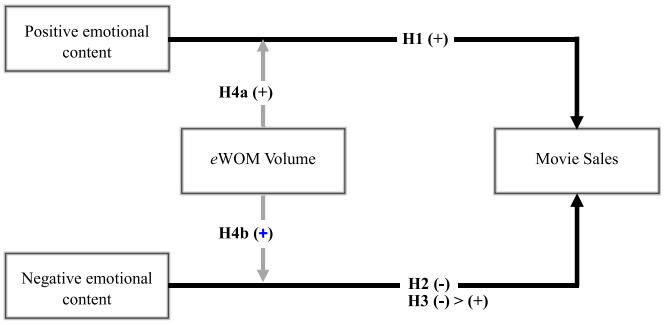


Fig. 1. Hypothesized model.

in *e*WOM (Fan et al., 2021; Robertson et al., 2023). Prior research shows that experience goods, particularly movies, generate strong emotional responses in consumers through *e*WOM (Liu, 2006; Robertson et al., 2023; Ullah et al., 2016). Given their short life cycle, with the majority of revenue generated within the initial weeks of release (Hennig-Thurau and Houston., 2019), movies are subject to extensive reviews by moviegoers (Fan et al., 2021; Robertson et al., 2023). Hence, the movie market is an exemplary domain for studying the causal effect of positive and negative emotional content on purchase likelihood and sales.

We obtained a list of Hollywood movies released between January 2011 and December 2011 from Box Office Mojo (Mojo), the online database that provides movies' financial information. The following criteria were used to select movies from the sampling frame representing different genres (i.e., Action, adventure, comedy, drama, and others). First, we shortlisted the most significant movies, i.e., those released nationwide in USA on more than 600 screens on the opening day, to ensure the market objective similarity of blockbusters. Nationwide released movies have broader appeal (being mainstream movies), generate *e*WOM, have rich availability of data, offer significant settings for cultural and commercial trends, and ensure standardized comparisons because they share a similar cinematic style (Chung et al., 2013; Gemser et al., 2007; Hadida, 2010). Second, given the transient nature of their revenue patterns (with about 90% generated during the initial release weeks), we selected movies for which the initial six-week revenue data was available (Liu, 2006). Third, movies with less than six weeks of reviews were excluded from the sample. Fourth, we removed movies for which no control variables data can be retrieved. Finally, we selected the year 2011 for two reasons: a) the industry conditions were restored from the global financial crisis and the strike of the Writers Guild of America[3] that significantly disrupted the film industry, and b) it was the year before the introduction of alternative movie platforms such as Netflix and Amazon Prime that impacted the movie distribution channels and shortened their life cycle. The final sample consists of 80 movies (names available on request) comprising some of the most significant blockbusters with emotional scenes and memorable characters that left their mark on consumer reviews.

3.2. Data and study variables

We collected data from authentic and publicly available sources, including Box Office Mojo (Mojo) and The Numbers—film industry data websites that systematically track movie sales, IMDb.com—a popular online database of film reviews and ratings with millions of registered users, and Rotten Tomato—a famous review-aggregation website for film. To compute our explanatory variables (positive and negative emotional content in *e*WOM), we compiled a sample of $N = 23{,}046$ USA consumer reviews written in English from IMDb. We extract the positive and negative emotional content from the online movie reviews by using Natural Language Processing (NLP). By applying the five-step NLP technique (Ullah et al., 2016) we were able to extract the most frequent emotional words, quantify them as the percentage of the total words, and classify positive and negative emotional content from each movie review. A list of the most frequently used emotional words is provided in Web Appendix A. Prior literature shows that *e*WOM often exhibits a "j-shaped distribution" of ratings with a majority of high scores, very few medium scores, and some low scores (Hu et al., 2017). Our data exhibit similar trend, i.e., the number of positive reviews (above average rating) remained significantly higher than that of negative reviews (below average rating). Therefore, we computed *e*WOM volume (moderator variable) as the total number of weekly reviews. We controlled for the confounding effects of variables that are considered necessary in the *e*WOM literature, e.g., average rating, *e*WOM variance, age, number of screens, production budget, star-power, movie genre, and critics' ratings (Carrillat et al., 2018; Kong and Lou, 2023; Palomba, 2020; Robertson et al., 2023). Table 1 describes all variables.

Table 1
Variables description and measures.

Variables	Description	Measures	Source
Sale (DV)	Weekly sales	Weekly movie sales in US$	Boxofficemojo.com
Positive emotion (IV	Percentage of positive emotional words	Weekly number of positive emotional words in reviews divided by total words in reviews	NLP Analysis
Negative emotion (IV)	Percentage of negative emotional words	Weekly number of negative emotional words in reviews divided by total words in reviews	NLP Analysis
Volume (M)	Weekly number of consumer reviews	Number of consumer reviews for each movie every week	IMDb.com
Ratings ©	Weekly average consumer ratings	Average movie ratings on a scale from 1 to 10, calculated every week	IMDb.com
Variance ©	The variance in user reviews	The standard deviation of user ratings for each movie every week	IMDb.com
Star power ©	The star power of actors featured in the movie	Ranking is based on the average movie sales from the previous three years for each star. A higher value indicates greater star power.	The-numbers.com
Critic ratings ©	Critics' reviews	Average ratings of critics on a scale of 1–10.	Rottentomatoes.com
Genre ©	Genre classification of the movies	Coded as three dummies, Action is the reference. Dummy-1: 1 = Drama; 0 = Action. Dummy-2: 1 = Comedy; 0 = action. Dummy-3: 1 = Other; 0 = Action	The-numbers.com, IMDb.com
Screen ©	Weekly number of screens	Number of screens on which a movie is being shown every week	Boxofficemojo.com
Budget ©	Production budget	The production budget in US dollars	Boxofficemojo.com
Age ©	Age of movie	Number of weeks that have passed since the movie's release	Boxofficemojo.com

Note: © Control variables; DV = Dependent variable; IV=Independent variable; M = Moderating variable.

[3] The 2007–08 Writers Guild of America (WGA) strike was a joint effort of two American labour unions representing writers in online media, film, radio, and television. The strike lasted from Nov 2007 to Feb 2008 and affected the production of several television shows and films.

Fig. 2 describes the NLP procedure and Table 2 reports the descriptive statistics.

3.3. Model

Our empirical model is based on the log formulation, which smoothens the distribution of variables in the regression model and deals with the skewness in data (Duan et al., 2008; Hansen et al., 2011). Previous research has shown that *e*WOM volume increases product sales (Chevalier and Mayzlin, 2016). However, at the same time, an increase in product sales may lead to an increase in *e*WOM activity, leading to reverse causality making the variable endogenous (Babić-Rosario et al., 2016). Additionally, previous research considers the number of screens endogenous because it is influenced by internal strategic decisions and external market demand (Clement et al., 2014). Nonetheless, valence, variance, and emotional content should not depend on the demand for movies but rather should be influenced by the experience of the consumers who write it (Hennig-Thurau and Houston, 2019), ruling out endogeneity concerns. Moreover, it is assumed that emotional *e*WOM content itself is not influenced by movie sales. While movie sales may have some impact on the volume of reviews, it is improbable for them to generate *e*WOM on specific emotions (Sanchez et al., 2020).

To deal with the issue of endogeneity, we used the instrument-free method introduced by Park and Gupta (2012), which significantly reduces the likelihood of endogeneity based on Gaussian copulas. We used the control function approach and added the Gaussian copula terms as regressors in the regression model (Park and Gupta, 2012). The copula term is a nonlinear transformation of the endogenous variables, which is the inverse normal cumulative distribution function $\Phi-1$ and the empirical cumulative distribution function H. Following Park and Gupta (2012): P*t = $\Phi-1$ (H(Pt)) where P*t is the copula term added to the model that estimates the correlation between the regressor and the error term scaled by the errors of variance. The following equation represents the primary model used for estimation:

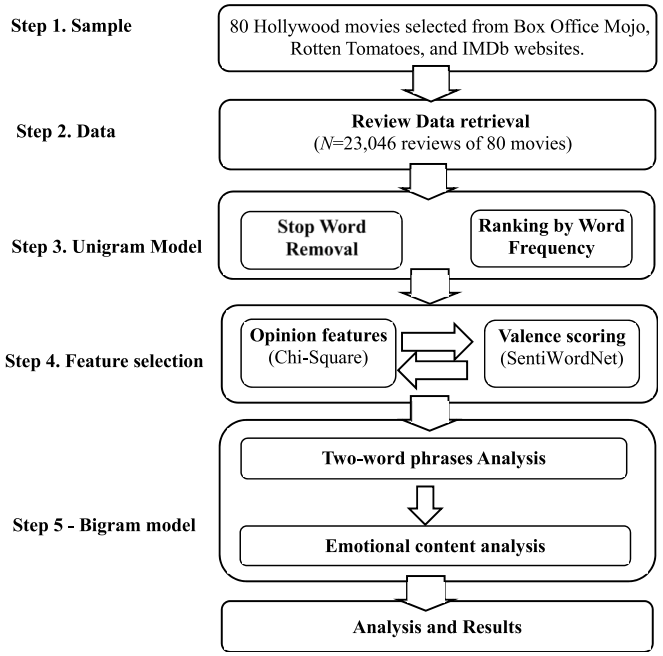


Fig. 2. Flowchart of NLP steps.

$$\log Sale_{it} = \alpha_0 + \alpha_1 \text{Pos_emotions}_{it} + \alpha_2 \text{Neg_emotions}_{it} + \alpha_3 \text{logVolume}_{it} + \alpha_4 \text{logRating}_{it} + \alpha_5 \text{logVarience}_{it} + \alpha_6 T_{it} + \alpha_7 Z_i + \alpha_8 D_i + \alpha_9 Pos_emotions_{it} * \text{logvolume}_{it} + \alpha_{10} Neg_emotions_{it} * \text{logvolume}_{it} + \alpha_{11} C_logVolume_{it} + \alpha_{12} \text{C_logScreens}_{it} + u_{it}$$

Where *t* represents each week of the initial six-week period of the sample ($t = 1, 2 \ldots 6$), *i* indexes the movies ($i = 1, 2 \ldots 80$), and u_{it} are the error terms representing all the unobserved influences on movie sales. The $Sale_{it}$ denotes the weekly sales for movie *i* at time *t* as a dependent variable. Pos_emotion and Neg_emotion represent the relative value of positive and negative emotional content. The logVolume, logRatings, and logVariance represent the *e*WOM control measures. The vector T_{it} consists of $Screens_{it}$ and Age_{it} time varying variables, whereas Z_i, consists of Stapower, and Critic ratings, and D_i represents Genre dummy categories. The Neg_emotion*logVolume and Pos_emotion * logvolume represent the interaction between the eWOM volume and positive and negative emotional content. While the C_logvolume and C_logScreen represent the copula term for *e*WOM volume and number of screens that control for the endogeneity in the regression model.

Following Park and Gupta (2012), the following equations represent the endogenous variables included in the model:

$$\text{C_logVolume}_{it} = \varphi^{-1}(H(\text{logVolume}_{it}))$$

$$\text{C_logScreens}_{it} = \varphi^{-1}(H(\text{logScreens}_{it}))$$

Where φ-1 is the inverse of the normal distribution function and H(.) is the empirical distribution of each regressor. The variables for Gaussian copula are non-normally distributed (Park and Gupta, 2012) and included after following the predetermined conditions required to include copulas (Becker et al., 2022). Moreover, for estimation, the log formulation of the continuous variables is used to get the coefficients except for the emotional content variables, which are ratios. The regression is estimated using bootstrap standard error with 500 repetitions. In addition, the variable inflation factor for all the variables was within the recommended limits, i.e., lesser than a conventional threshold of 10, indicating that multicollinearity is not an issue for the analysis (Kumar and Zaheer, 2019). We also run several robustness checks of the estimated model (see Web Appendix B).

4. Results

We test six models by adding variables to a previous model. Model 1 was run for control variables; Model 2 comprises the main effects of positive and negative emotional content; Model 3 includes Copula; and Model 4 examines the moderation effect of total *e*WOM volume. Additionally, for our post-hoc analysis, Model 5 evaluates the discrete impact of positive versus negative eWOM volume, and Model 6 examines their moderation effects. This systematic procedure allowed us to examine if each of our theorized constructs significantly improved movie sale's R^2 after accounting for the previous predictor(s).

4.1. Main effects

In Model 1, the direct effects of total *e*WOM volume and the control variables jointly explain 61.9 % of the variance in movie sales. Results show that the direct effect of volume is positive and significant ($\beta = 0.160$, $p < 0.01$). The significant direct effect of volume holds in the subsequent emotional content and interaction models (Model 2: $\beta = 0.111$, $p < 0.01$; Model 3: $\beta = 0.79$, $p < 0.1$ and Model 4: $\beta = 0.188$, $p < 0.01$). As predicted, some control variables also significantly affected movie sales. Results are provided in Table 3.

Model 2 shows a significant positive relationship between positive emotional content and movie sales ($\beta = 0.224$, $p < 0.01$). Model 3 with

Table 2
Summary statistics.

Variables	Mean	SD	Min	Max	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1. Sale[@]	15.20	22.10	0.021	226.0										
2. Volume	25.08	34.22	1	272	0.51*									
3. Ratings	6.28	1.56	0	9.66	0.21*	0.13*								
4. Variance	6.36	4.44	0	27	0.04	0.11*	−0.36*							
5. Screen	2336.94	1102.37	44	4375	0.70*	0.54*	0.05	0.00						
6. Star power	142.92	88.36	1	369	0.32*	0.05	0.19*	−0.02	0.21*					
7. Budget[@]	71.50	55.40	1.50	250.0	0.33*	0.14*	0.13*	0.01	0.21*	0.37*				
8. Age	9.285	1.191	5.4	12.36	−0.30*	−0.25*	−0.03	−0.07	−0.19*	−0.12*	−0.11*			
9. Critic	57.05	27.58	5	96	0.16*	0.08	0.33*	−0.15*	0.09*	−0.09*	0.03	0.07		
10. Positive emotions	11.65	8.57	0.43	86.92	0.22*	0.17*	0.03	0.02	0.11*	0.05	0.03	−0.24*	−0.11*	
11. Negative emotions	4.32	4.36	0	41.66	−0.25*	−0.19*	−0.15*	0.01	−0.14*	−0.01	−0.11*	0.10*	−0.14*	0.30*

Note: [@] in Million US$; Numbers rounded off to three decimal figures.

Table 3
Regression.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	4.637*** (1.26)	4.090*** (1.240)	6.904*** (1.988)	7.083*** (2.137)	4.321*** (1.456)	4.504*** (1.459)
Total *e*WOM volume	0.160*** (0.046)	0.111*** (0.046)	0.079* (0.049)	0.188*** (0.066)		
Controls						
Rating	0.718*** (0.025)	0.581*** (0.220)	0.572*** (0.221)	0.513** (0.229)	0.693*** (0.250)	0.647*** (0.252)
Variance	0.146*** (0.053)	0.134*** (0.052)	0.124*** (0.052)	0.120*** (0.047)	0.132*** (0.062)	0.129** (0.061)
Screens	0.993*** (0.068)	0.995*** (0.066)	0.800*** (0.118)	0.744*** (0.140)	1.058*** (0.076	1.026*** (0.078)
Star power	0.111*** (0.047)	0.117*** (0.046)	0.124*** (0.046)	0.120*** (0.045)	0.067 (0.053)	0.064 (0.053)
Budget	0.213*** (0.053)	0.204*** (0.051)	0.186*** (0.052)	0.192*** (0.054)	0.188*** (0.058)	0.185*** (0.058)
Comedy	0.267** (0.127)	0.185* (0.126)	0.160 (0.128)	0.179* (0.112)	0.089 (0.142)	0.119 (0.144)
Drama	0.424** (0.135)	0.402*** (0.133)	0.397** (0.133)	0.395*** (0.341)	0.319*** (0.154)	0.310** (0.155)
Other	−0.084 (0.125)	−0.048 (0.121)	−0.065 (0.121)	−0.053 (0.063)	−0.150 (0.138)	−0.142 (0.137)
Age	−1.499*** (0.337)	−1.036*** (0.340)	−0.987*** (0.342)	−0.956*** (0.609)	−1.240*** (0.039)	−1.300*** (0.398)
Critic ratings	0.158*** (0.067)	0.153*** (0.066)	0.156*** (0.066)	0.185*** (1.597)	0.195*** (1.456)	0.209*** (0.078)
Main effect						
Positive emotions		0.022*** (0.00547)	00.022*** (0.00548)	0.066*** (0.024)	1.407*** (0.599)	3.642*** (1.52)
Negative emotions		−0.047*** (0.011)	−0.048*** (0.011)	−.0.090** (0.041)	−5.411*** (1.266)	−5.451*** (1.634)
Positive volume					0.144*** (0.065)	0.243*** (0.090)
Negative volume					0.043 (0.071)	0.051 (0.087)
Interaction effects						
Positive emotions x total volume				0.017*** (0.006)		
Negative emotions x total volume				0.025* (0.015)		
Positive emotions x positive volume						0.779* (0.410)
Negative emotions x negative volume						−0.236 (1.220)
Screens-Copula			0.194** (0.099)	0.211*** (0.107)		
Volume-Copula			0.449 (0.599)	0.559* (0.349)		
Positive emotions x volume-Copula				0.107 (0.187)		
Negative emotions x volume-Copula				−0.178 (0.136)		
Adjusted R^2	0.619	0.631	0.632	0.639	0.604	0.607
RMSE	0.876	0.852	0.851	0.843	0.588	0.589
W		2.14***	4.061***	0.492***		

Note: ***, **, and * Mean difference is significant at the ≤0.01, ≤0.05 and ≤0.1 levels, respectively. Standardized regression β weights are shown with Standard error in parentheses. Dependent Variable = Movie weekly sales, RMSE = Root mean square error; *W*=Wald test.

endogeneity control confirms these results ($\beta = 0.221$, $p < 0.01$ and $\beta = 0.667$, $p < 0.01$) in support of H1. Model 2 results further show that negative emotional content has a significant negative impact on movie sales ($\beta = -0.479$, $p < 0.01$) in support of H2. The results are replicated in Model 3 with endogeneity control ($\beta = -0.482$, $p < 0.01$ as well as in Model 4 ($\beta = -0.901$, $p < 0.05$). To check the negativity bias, i.e., whether the effect of negative emotional content on movie sales is more pronounced than the positive emotional content, we performed the Wald test for all the variables included in Model 2, Model 3, and Model 4, respectively. The results of the Wald test (Model 2: $W = 2.14$, $p < 0.01$; Model 3: $W = 4.061$, $p < 0.01$, and Model 4: $W = 0.492$, $p < 0.01$) suggest that the effects of negative emotional content are significantly greater than that of positive emotional content thus confirming H3.

4.2. Moderating effect

Model 4 was run to examine the moderating effect via the interaction of total *e*WOM volume and emotional content in *e*WOM on movie sales after controlling for endogeneity.[4] Results indicate that positive emotional content's interaction with *e*WOM volume is positive and significant ($\beta = 0.168$, $p < 0.01$), supporting H4a. That is, the positive effect of emotional content on movie sales becomes stronger with higher *e*WOM volume. Model 4 further shows that the interaction of negative emotional content with *e*WOM volume is also positive and marginally significant ($\beta = 0.249$, $p < 0.1$) in support of H4b. In other words, when the total *e*WOM volume increases, the negative effect of negative emotional content on movie sales weakens. To visually present the hypothesized moderating relationships and results, we used the

[4] The significant effect of the copula term of the screen and WOM volume (Model 3: $\beta = 0.194$, $p < 0.01$ and $\beta = 0.549$, $p < 0.10$) confirms the endogeneity and supports the model with endogenous control copula terms (Park and Gupta, 2012).

unstandardized β coefficients to plot the interaction effects using simple slope test (Aiken et al., 1991). The test involved splitting the moderators into a low group (one standard deviation below the mean) and a high group (one standard deviation above the mean) and re-estimating the relationship by using the range of 10th to 90th percentile for both positive and negative emotion variables. The plot on the left in Fig. 3 shows that when the *e*WOM volume is high, the positive relationship between positive emotional content and movie sales is stronger than when the *e*WOM volume is low. The plot on the right shows that when *e*WOM volume is high, the negative relationship between negative emotional content and movie sales is stronger than when the *e*WOM volume is low. However, as is evident from the figure, the slope gradient for higher *e*WOM volume is significantly flatter than the slope gradient for lower *e*WOM volume, which corroborates the diminishing returns postulated in H4b.

4.3. Post-hoc analysis

To gain further insight and measure the discrete impact of positive and negative *e*WOM volume on movie sales, we calculated positive and negative volume. Positive volume is measured as the total number of positive reviews, while negative volume is the total number of negative reviews (Babic-Rosario et al., 2016). We estimated Model 5 to analyze the positive and negative volumes' direct impact and Model 6 for their moderation effects. We aim to examine if our hypothesis H4b holds with increasing volume of all positive versus all negative reviews. Model 5 shows that positive volume has positive and significant impact on movie sales ($\beta = 0.144$, $p < 0.01$), consistent with model 3. However, the negative volume shows an insignificant impact on movie sales ($\beta = 0.043$, ns). Furthermore, Model 6 shows that the interaction effect of higher positive volume and emotional content on movie sales is positive and marginally significant ($\beta = 0.779$, $p = 0.06$), indicating similar results to Model 4. Whereas the interaction between higher negative volume and emotional content on movie sales remains insignificant ($\beta = -0.236$, ns), showing that an increasing *e*WOM having all negative reviews has no moderating effect on the relationship between negative emotional content and movie sales.

Our results of *e*WOM emotional content and volume can be used to predict movie sales. By crossing these two dimensions (volums and valence), we can yield a 2 X 2 matrix that locates four categories used to describe movie sales: Flop, Hype, Star, and Hit (see Fig. 4).

Quadrant 1 depicts a 'Flop' movie that elicits negative but a lower volume of consumer reviews within six weeks of release. The initial negative *e*WOM spirals customers' interest in the movie, decreasing purchase likelihood and reducing movie sales. Given little interest by moviegoers, it is conceivable that the flop movie will have a short life span, and the movie sales will decline. Quadrant 2 represents a 'Hype' movie, attracting higher consumer reviews with negative emotional content within six weeks of release. As our data show, when negative

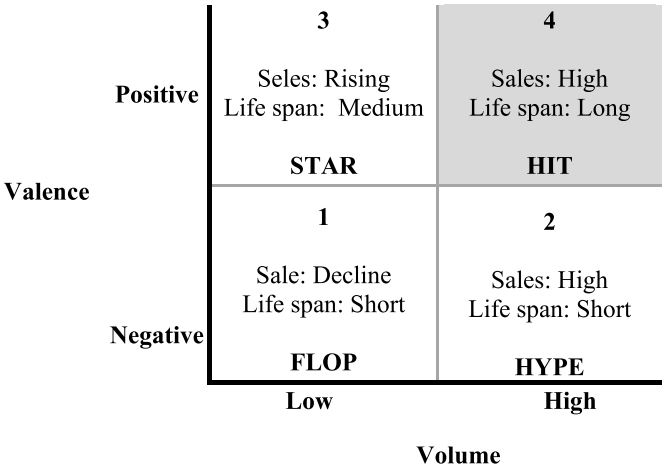


Fig. 4. Emotional content and movie sales.

*e*WOM increases beyond a certain limit, a balancing loop comes into play. A mechanism that counters change in one direction with a change in the opposite direction and seeks to stabilize a system (Senge, 1990). The results of curvilinear regression analysis also confirm this phenomenon by showing a negative and significant relationship between negative emotional content and movie sales ($\beta = -0.530$, $p < 0.05$). Moreover, the effect of the squared term of negative emotional content on movie sales is positive ($\beta = 0.124$), suggesting that the negative

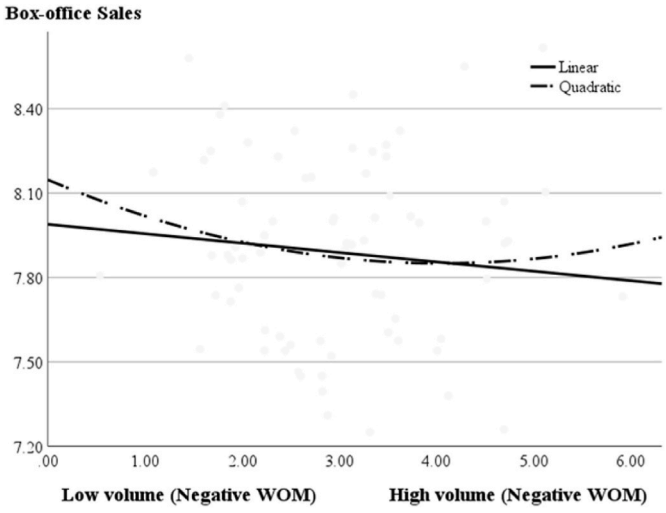


Fig. 5. Results of curve estimation.

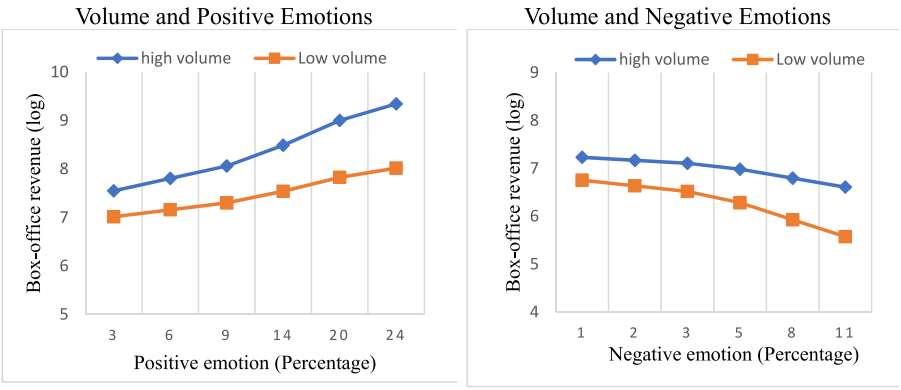


Fig. 3. Simple slope test depicting the interaction effects.

impact on movie sales is diminishing (Fig. 5). That is, when negative *e*WOM volume increases to higher levels, the detrimental effects of negative content start to diminish, and movie sales increase. As such, bad publicity and controversy attract attention, resulting in customer excitement and interest that sells out movies instantly, and thus the adverse effect of negative emotional content on movie sales is attenuated.

Quadrant 3 depicts a 'Star' movie, one that is getting significant attention from moviegoers. It attracts some positive *e*WOM within six weeks of release. These initial positive reviews enhance the purchase likelihood and movie sales in the long run. Given positive reviews, it is reasonable to predict that a Star movie will have a longer life span and the sales are likely to rise. Finally, Quadrant 4 illustrates a 'Hit' movie, one which has attracted a higher volume of consumer reviews with positive emotional content within the first six weeks. Given the rising interest by moviegoers, the Hit movie will have a longer life span and will likely generate greater revenue.

5. Discussion

5.1. Conclusions

The valence and volume of *e*WOM content can significantly influence consumers' purchase decisions and sales. Prior studies have explored this phenomenon, primarily focusing on positive and negative sentiments. However, positive and negative emotional content, which greatly drives *e*WOM, remained less explored. Since emotions significantly differ from sentiments in their characteristics and influence on consumer behavior, studying their specific impact on sales is crucial. This study analyzed 23,046 reviews of 80 Hollywood movies for positive and negative emotional content and empirically examined their impact on movie sales, using *e*WOM volume as a moderating variable. We find that: a) positive emotional content enhances and negative emotional content reduces movie sales, the later has a more substantial effect, and b) with the increasing *e*WOM volume, the effect of positive content increases progressively, whereas negative emotional content exhibits diminishing returns, i.e., beyond a certain limit, an increase in *e*WOM volume with negative emotional content has a positive impact on movie sales. We used emotional content valence and volume as precursors to derive a 2x2 matrix that can broadly anticipate four categories of movie sales—Flop, Hype, Star, and Hit. Our research expands the understanding of how emotional content in eWOM influences moviegoers' choice of movie and impacts sales. The findings have significant theoretical and managerial implications:

5.2. Theoretical implications

This research contributes to several literature streams. First, we extend the applications of EASI theory (Alkhadher et al., 2023; Deng et al., 2020; Van Kleef and De Dreu, 2010; Van Kleef, 2017) to the context of film industry by examining the social influence of online movie reviews. Using NLP technique, our research examined positive and negative emotions embedded in *e*WOM content as the drivers of social influence and movie sales, adding a new perspective to the existing EASI theory. Methodologically, the NLP extraction algorithms more precisely measured the proportion of emotional content in online reviews, allowing us the examination of previously unexplored characteristics of emotional content in *e*WOM, since prior studies in consumer psychology have shown that emotions play a significant role in consumer decision making (Lin et al., 2018; Shiv et al., 2005; Verhulst et al., 2020). Second, by distinctively examining the effects of positive and negative emotional content on movie sales, we contribute to the growing literature on *e*WOM (Babic Rosario et al., 2016; Basuroy et al., 2020; Berger and Milkman, 2012; Hennig-Thurau et al., 2006; Ullah et al., 2016). The findings support recent theoretical developments in the field. For example, our results endorse Brady et al. (2023, p. 918) assertion about the "crowd-emotion-amplification effect"—customers over-perceive the extremity of group emotions when they are exposed to emotional reviews of many group members at once. Our data also confirm the notion of "negativity bias", i.e., consumers value negative information more than positive information (Robertson et al., 2023). Our study significantly extends this stream by demonstrating the moderating role of *e*WOM volume, which has been overlooked in previous studies. We found that with increasing *e*WOM volume, the effect of positive emotional content on movie sales strengthens, whereas negative emotional content has a curvilinear effect. It increases initially; however, it diminishes at higher *e*WOM volume.

Additionally, we employed two conceptual distinctions (emotional content valence and emotional content volume) discussed in previous *e*WOM literature to develop a 2 X 2 matrix. The resulting typology—Flop, Hype, Star, and Hit—contributes valuable extensions to previous studies that examined the *e*WOM relationship with movie performance (Babic-Rosario et al., 2016; Liu, 2006; You et al., 2015; Rocklage and Fazio, 2020). The matrix can be used to anticipate movie sales by plotting positive and negative emotions vis-a-vis review volume. Finally, we contribute to the entertainment industry literature that explores the drivers of movies box-office performance. Previous studies demonstrate that release timing, distribution, awards received, critic reviews, and *e*WOM drive a movie's box-office performance (Basuroy et al., 2003; Hennig-Thurau et al., 2006; Hennig-Thurau and Houston, 2019). Our research extends this literature by showcasing how positive and negative emotional content in *e*WOM impacts box-office sales after the movie's release.

5.3. Managerial implications

This study provided preliminary steps towards a deeper understanding of online review content, which is far more sophisticated compared to the user ratings typically used. Given our findings marketers, studio managers, and movie distributors may incentivize consumers to share more positive reviews online to increase movie sales. This underscores the necessity for a more sophisticated approach to managing online reviews, beyond aggregate *e*WOM metrics. This strategy is equally helpful for 'Flop' and 'Star' movies presented in quadrants 1 and 3. Strategies such as engaging with online communities through post-movie surveys, offering rewards for feedback, and conducting targeted social media campaigns encourage the dissemination of positive emotional experiences. These efforts can cultivate a favorable movie perception and will contribute to increased box-office performance. Furthermore, our results demonstrated that at high *e*WOM volume, the effect of negative content weakens. As such, high negative publicity increases product awareness, evokes curiosity, and fosters customer excitement (Berger et al., 2010; Mohr, 2007; Reichstein and Brusch, 2019). Moreover, the bandwagon effect (as discussed earlier) can persuade potential moviegoers' preference for the movie and influence their intent to buy (Meeker, 2022). Therefore, managers can use the product's popularity to weaken the effect of negative *e*WOM.

While our results support previous studies that emotional appeals influence consumers' decision-making (Archer, 2016; Yi et al., 2023), we recommend online retailers to analyze movie reviews, specifically focusing on emotional content, to better understand moviegoers' emotional contagion. Appendix A provides a library of example emotional words that can be used to analyze online review items. Our findings also hold significant value for manufacturers and marketers from the advertising perspective. They can choose which emotional expressions to incorporate into movie promotions. For instance, phrases like "lovely story", "amazing video", and "great action", which emerged prominently in our data, could be emphasized in advertisements. Previous research shows that emotionally charged content fosters greater sharing, increasing page views, as corroborated by prior research (Berger and Milkman, 2012; Goldenberg and Willer, 2023). Additionally, these emotional cues serve as relevant descriptors for the movie,

facilitating consumers' cognitive processing of information in the digital marketplace. This aligns with research in neuroscience and psychology (Lin et al., 2018; Shiv et al., 2005; Verhulst et al., 2020), affirming the positive and vital role emotions play in consumer decision-making.

5.4. Limitations and further research

Although this study provides valuable insights, it has some limitations that serve as signposts for future research. From the data perspective, we did not calculate the exact revenues and their time horizon for Flop, Hype, Star, and Hit movies. Future quantitative studies should take this into account for more accurate predictions about movie sales and the life span of each category. Moreover, our sample of 80 movies from 2011 may limit the generalizability of our findings. Given the insignificant direct and moderating impact of increasing negative eWOM volume on movie sales in this study, we recommend that future research expand the data domain to include more movies and other forms of user-generated content, such as tweets, blogs, and spoilers, to study the patterns of *e*WOM more deeply. Theoretically, we are constrained by our use of movie sales as the dependent variable, as aftermarket sales (e.g., DVDs) were not considered. We call for future studies to use the EASI lens to explore aftermarket release reviews that may uncover new patterns of *e*WOM. Finally, from the method perspective, we relied on NLP, which has significant accuracy but is still not equivalent to human-level comprehension. Combined with our methods, future advancements are expected to yield even more reliable results.

CRediT authorship contribution statement

Rahat Ullah: Writing – review & editing, Project administration, Conceptualization. **Muhammad Aftab Alam:** Writing – review & editing, Writing – original draft, Project administration, Data curation. **Atya Zeb:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation.

Declaration of competing interest

The authors have had no competing interests to declare. No Funding was received.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104304.

Data availability

Data will be made available on request.

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